

ST ALBERT, AB, Canada

WMO: 711215

Lat: 53.6920N

Lon: 113.6196W

Elev: 687

StdP: 93.34

Time Zone: -7.00 (NAM)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-31.0	-27.6	-34.4	0.2	-30.9	-30.8	0.2	-27.3	10.8	-4.2	9.2	-4.6	1.8	110	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	27.6	17.1	26.1	16.6	24.6	16.1	19.4	25.1	18.2	23.9	17.2	22.8	3.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	13.3	23.1	15.9	12.3	21.3	14.9	11.5	20.1	58.3	24.9	54.3	23.9	50.9	22.7	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.6	7.5	DB	-36.0	31.1	3.3	1.6	-38.4	32.2	-40.3	33.1	-42.1	34.0	-44.5	35.1	(4)
(5)			WB	-36.1	21.6	3.3	1.5	-38.5	22.7	-40.4	23.6	-42.2	24.4	-44.6	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-10.0	-11.3	-5.1	3.2	11.4	14.8	16.7	15.6	10.7	3.8	-5.6	-10.9	(6)
(7)		DBStd	12.05	8.34	8.80	7.98	5.34	4.42	2.93	2.61	2.94	4.44	4.51	7.14	7.37	(7)
(8)		HDD10.0	3290	619	597	468	209	37	3	0	1	44	197	467	647	(8)
(9)		HDD18.3	5677	878	831	727	453	217	113	63	92	229	452	717	906	(9)
(10)		CDD10.0	683	0	0	0	6	79	145	209	176	65	4	0	0	(10)
(11)		CDD18.3	30	0	0	0	0	2	6	14	8	1	0	0	0	(11)
(12)		CDH23.3	615	0	0	0	2	87	90	214	157	63	1	0	0	(12)
(13)		CDH26.7	89	0	0	0	0	13	9	35	20	12	0	0	0	(13)
(14)	Wind	WSAvg	3.4	3.5	3.4	3.3	3.9	4.0	3.8	3.1	2.9	3.2	3.6	3.1	3.2	(14)
(15)	Precipitation	PrecAvg	428	19	11	16	28	50	75	87	57	32	20	19	15	(15)
(16)		PrecMax	596	72	28	33	49	102	154	142	121	82	57	71	25	(16)
(17)		PrecMin	242	3	0	5	3	10	16	21	22	5	5	4	5	(17)
(18)		PrecStd	82	16	6	8	12	22	35	31	25	22	12	15	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	6.9	12.0	22.7	28.6	27.9	29.9	28.8	28.5	20.1	11.8	5.6	(19)
(20)			MCWB	2.8	3.5	5.6	10.9	13.7	16.1	20.0	17.5	16.0	11.4	6.4	2.3	(20)
(21)		2%	DB	4.9	4.2	8.8	17.9	26.0	25.9	27.4	26.6	25.0	16.4	7.8	3.4	(21)
(22)			MCWB	2.1	1.4	4.3	8.4	12.6	16.0	18.9	17.9	14.7	8.6	4.0	0.8	(22)
(23)		5%	DB	3.4	2.6	6.4	14.9	23.3	23.9	25.7	25.2	22.5	14.0	5.4	1.5	(23)
(24)			MCWB	1.1	0.3	2.9	6.9	11.8	14.9	18.0	17.5	14.0	7.8	2.8	-0.7	(24)
(25)		10%	DB	1.6	0.8	4.3	12.2	20.9	22.1	24.0	23.5	19.5	11.5	3.2	-0.8	(25)
(26)			MCWB	-0.2	-0.9	1.5	5.7	10.7	14.2	17.1	16.5	12.6	6.4	1.4	-2.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.6	6.7	10.9	14.6	18.7	21.4	20.6	17.2	11.5	7.2	2.6	(27)
(28)			MCD B	5.8	6.8	10.7	21.9	24.5	24.9	28.0	25.9	25.5	19.5	10.8	5.2	(28)
(29)		2%	WB	2.2	1.7	4.5	8.7	13.4	17.0	19.8	19.2	15.6	9.3	4.7	0.9	(29)
(30)			MCD B	4.6	3.7	8.4	17.2	23.1	23.6	25.7	24.8	23.6	15.2	7.4	3.1	(30)
(31)		5%	WB	1.2	0.5	3.0	7.2	12.5	15.9	18.7	17.9	14.2	8.1	2.9	-0.6	(31)
(32)			MCD B	3.2	2.3	6.1	14.0	21.7	21.8	24.4	23.5	21.7	12.5	4.8	1.6	(32)
(33)		10%	WB	-0.2	-0.8	1.6	5.9	11.4	15.0	17.6	16.9	13.0	6.9	1.5	-2.5	(33)
(34)			MCD B	1.6	0.9	3.9	11.5	19.8	20.8	23.0	22.5	18.7	10.9	3.3	-0.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.8	10.8	10.6	10.9	13.8	11.9	12.3	13.4	13.3	11.5	8.7	9.2	(35)
(36)			MCDBR	9.2	10.9	12.6	16.4	17.8	15.1	15.6	16.0	18.8	15.6	10.7	10.6	(36)
(37)			MCWBR	7.2	8.5	8.5	8.5	7.7	6.9	8.3	8.4	9.5	8.9	7.5	8.6	(37)
(38)		5% WB	MCDBR	9.3	10.5	11.9	15.2	16.2	13.2	14.3	14.8	17.6	14.2	9.3	10.6	(38)
(39)			MCWBR	7.5	8.4	8.0	8.1	7.3	6.6	8.2	8.3	9.1	8.4	6.8	8.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.255	0.267	0.310	0.326	0.341	0.351	0.354	0.360	0.307	0.281	0.253	0.238	(40)	
(41)		taud	2.308	2.352	2.307	2.392	2.395	2.382	2.420	2.352	2.500	2.549	2.456	2.319	(41)	
(42)		Ebn at Noon	733	849	873	907	906	898	887	859	875	832	748	685	(42)	
(43)		Edh at Noon	56	75	99	104	110	113	107	108	81	62	49	45	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.95	1.99	3.52	4.67	5.70	6.00	6.12	5.04	3.48	2.00	0.95	0.70	(44)	
(45)		RadStd	0.10	0.15	0.27	0.43	0.43	0.39	0.31	0.35	0.36	0.24	0.11	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page